Sunday School Scheduler — Design Document

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Reviewers	[who reviewed it — a teammate, the instructor, a design-review partner]
Status	In review
Last updated	2026-10-21
Links	Proposal · Repository · Issue tracker

Revision history

Details for each revision are in the commit messages (`git log docs/design.md`).

Date	Author	What changed
2026-10-16	Hyrum Kammerman	First draft
2026-10-20	Hyrum Kammerman	Revised after draft review; Mermaid diagrams
2026-10-20	Hyrum Kammerman	Moved the services to a Node backend on EC2; trimmed to length
2026-10-21	Hyrum Kammerman	Trimmed a bit more and completed final proofread before submission.

1. Context and scope

Sunday School Scheduler is a web app that helps a Sunday School presidency in The Church of Jesus Christ of Latter-day Saints make sure every class on every Sunday has a teacher. Today my ward schedules teachers in a spreadsheet and confirms them by copying and pasting individual messages. The app shows the presidency which lessons are covered, gives teachers a frictionless way to confirm or ask for a substitute, and emails people when assignments are made or answered.

2. Goals and non-goals

Goals

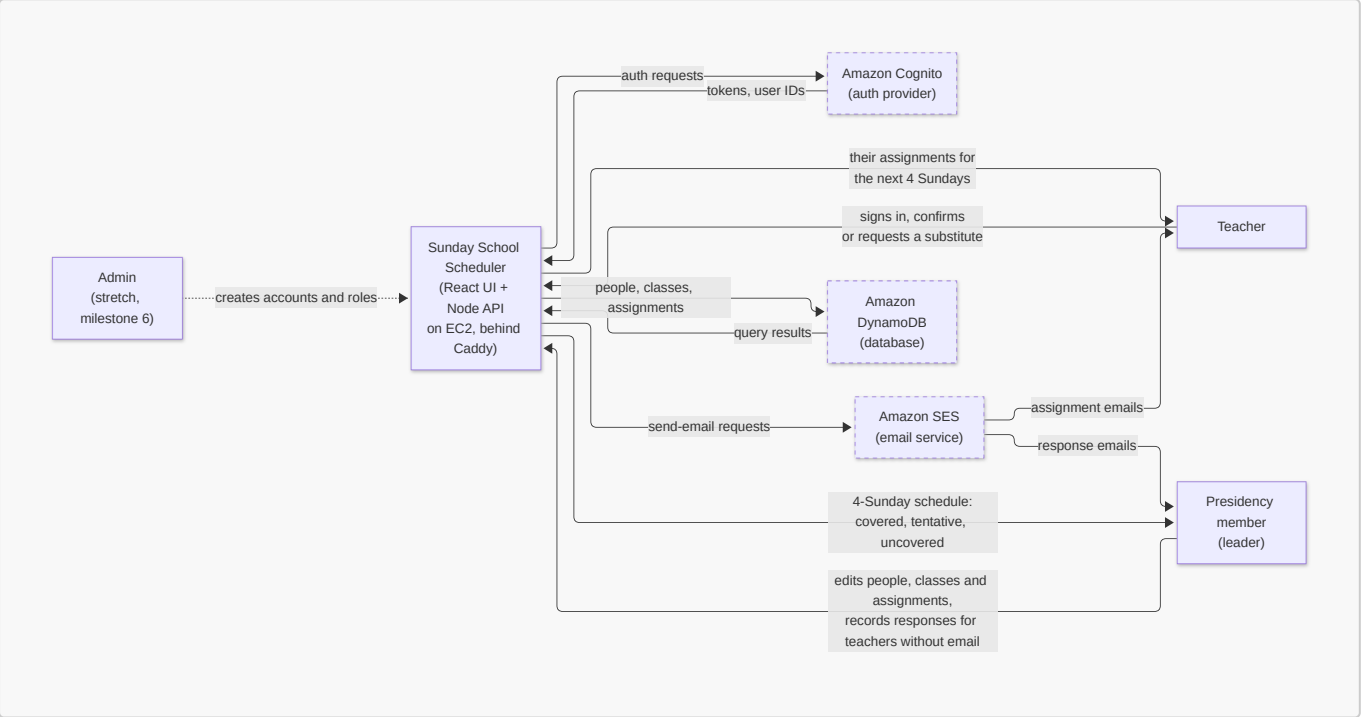
- A presidency member never has to copy and paste messages to teachers: assigning or reassigning a teacher emails that teacher automatically within 5 minutes.
- A teacher can see their assignments within the next 4 Sundays and confirm a lesson or request a substitute in under 1 minute.
- When a teacher requests a change, the presidency is notified within 5 minutes.
- A presidency member can see who is assigned to every class for the next 4 Sundays in under 1 minute, and spot the uncovered classes at a glance.

- Running cost stays near zero, within free tiers wherever possible.

Non-goals

- Other organizations (Relief Society, Elders' Quorum, Primary). The data model is Sunday School–specific.
- More than one ward. There is no multi-tenant separation in the MVP.
- Lesson-plan and content sharing between teachers.
- SMS notifications. Email only for the MVP (see §6).
- Importing members from Member Tools or other Church systems. People sign up, and leader roles are added by hand until admin functionality is implemented.

3. System-context diagram



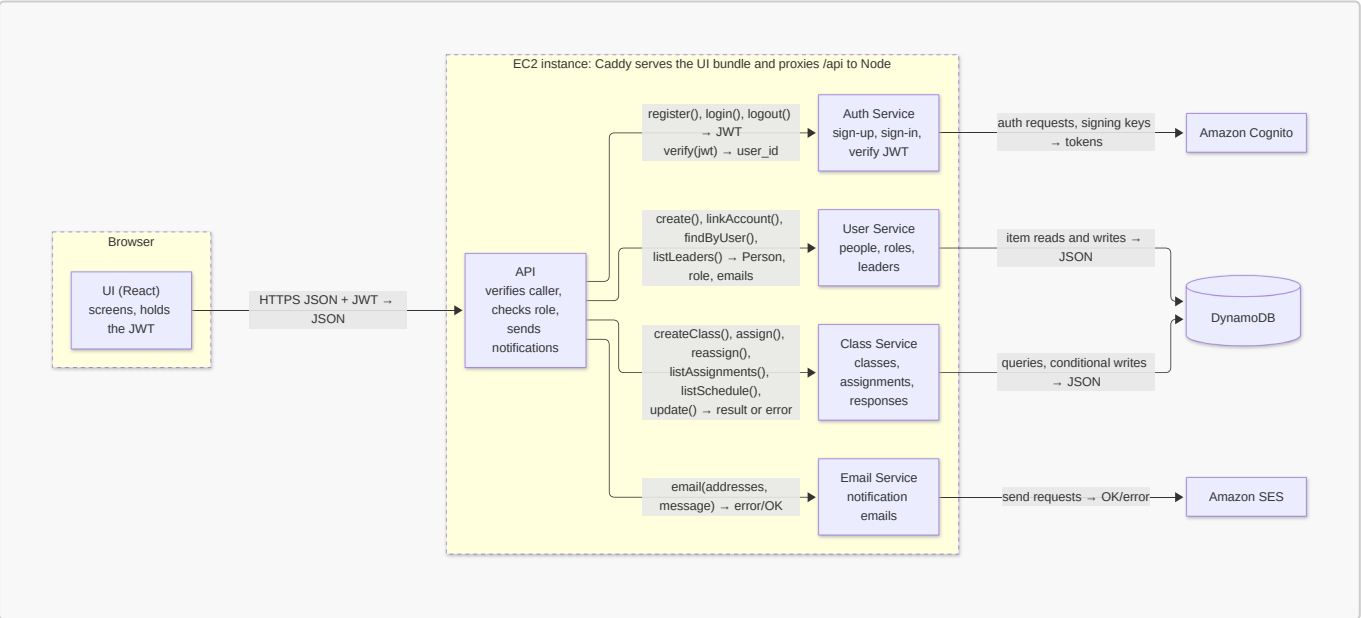
4. Architecture and components

The UI runs in the browser. Everything else is one Node (TypeScript) process on an EC2 instance: an API that every request passes through, and four services behind it. The browser never holds AWS credentials; only the instance's IAM role can call DynamoDB and SES.

Component	Responsibility	Talks to	Interface (rough)
UI	Sign-in, teacher dashboard, leader schedule, teacher actions, and messages.	API	HTTPS JSON with the JWT, e.g. `POST /login`, `GET /assignments`, `PUT /assignments/{class}`

Component	Responsibility	Talks to	Interface (rough)
API	Verifies caller, checks their role, calls services, and sends notifications.	UI and the four services	HTTP routes returning JSON and status codes (200, 403, 500)
Auth Service	Signs users up and in through Cognito and verifies JWTs	Cognito	`register()`, `login()`, `logout()` → JWT; `verify(jwt)` → user_id
User Service	Creates people, links and finds accounts and roles	DB	`create(name, email?)`, `linkAccount(user_id, email)`, `findByUser(user_id)` → Person, `listLeaders()` → emails
Class Service	Classes, assignments and responses	DB	`createClass()`, `assign(class, person)`, `reassign(class, from, to)`, `listAssignments(person, dates)`, `listSchedule(dates)`, `update(class, person, status)`
Email Service	Sends notification emails.	SES	`email(addresses, message)` → OK/error

Who can do what. A teacher sees and responds to only their own assignments; anything else returns 403. A leader can also create classes and people, assign and reassign teachers, record responses for anyone, and see the full schedule. An admin manages accounts and roles (stretch milestone 6).



Hosting

One EC2 instance, in the same AWS account as Cognito, DynamoDB and SES, runs Caddy in front of the Node process. Caddy serves the built UI bundle, provides HTTPS on the app's domain with automatic certificates, and forwards `/api` requests to Node. The instance's IAM role grants access to the app's tables

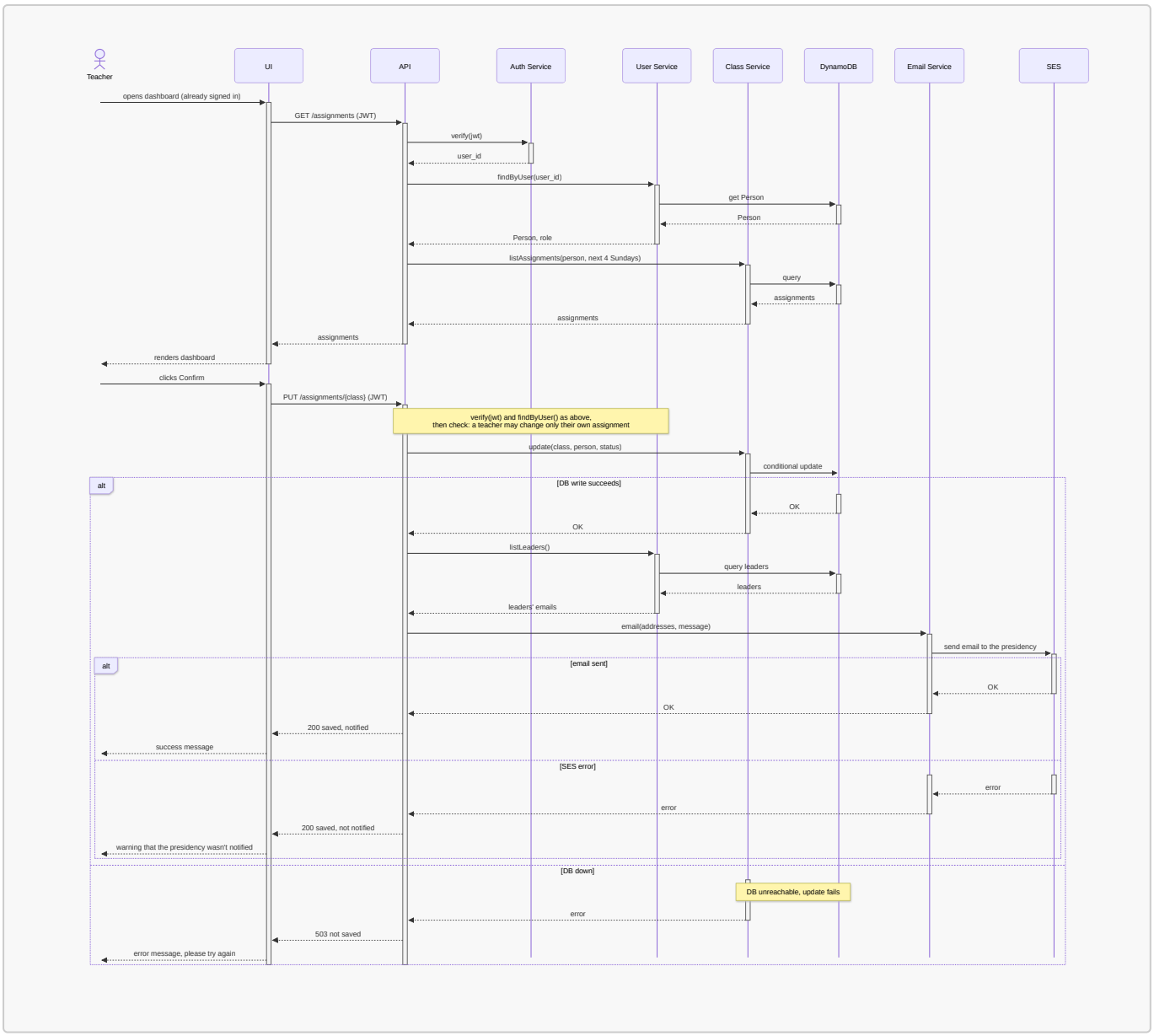
and to SES, so no AWS keys live in the code. The instance holds no data, so it can be rebuilt from the repository; the setup steps belong to CP7.

Core flow, traced end to end

A teacher confirms an assigned lesson.

1. The signed-in teacher opens the dashboard.
2. UI → API `GET /assignments`. The API identifies the caller with `verify(jwt)` → user_id and `findByUser(user_id)` → Person and role.
3. API → Class Service `listAssignments(person, next 4 Sundays)`. The UI renders the dashboard.
4. The teacher clicks **Confirm**. UI → API `PUT /assignments/{class}`. The API identifies the caller again and checks that the assignment is theirs (or that they're a leader).
5. API → Class Service `update(class, person, status)`, a conditional write that returns OK.
6. API → User Service `listLeaders()`, then Email Service `email(addresses, message)` through SES.
7. The API returns success and the UI shows a success message. If SES fails after the save, the API still returns 200 and the UI warns that the presidency wasn't notified.

A substitute request is the same flow with a different `status`.



The failure case

The database is down when the teacher clicks Confirm. The conditional write fails, Class Service returns an error, and the API returns 503 without calling the Email Service. The UI says "Couldn't save your response, please try again." Nothing changed and nobody was emailed, and saving the same status again is safe, so recovery is simply retrying.

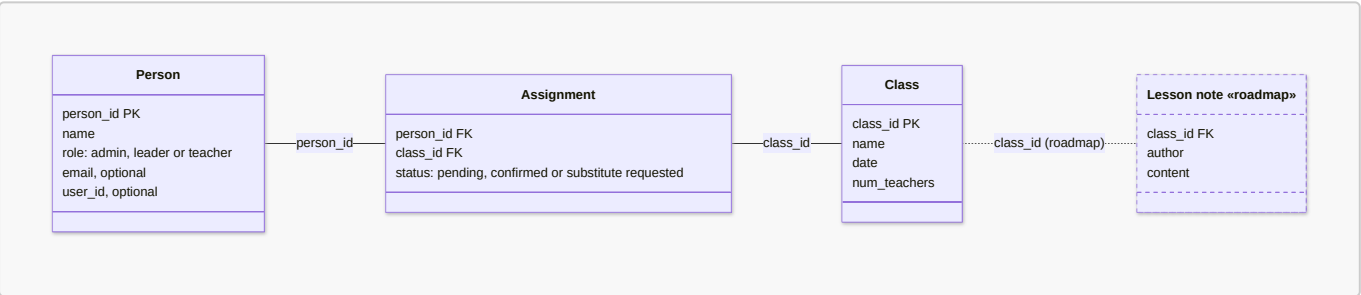
5. Data model

Entity	Key fields	Relationships	Owned by	MVP or roadmap
Person	person_id, name, role (admin / leader / teacher), email (optional), user_id (optional)	Has many Assignments; may link to one Cognito account	User Service	MVP
Class	class_id, name, date, num_teachers	One meeting of a class on one Sunday; has many Assignments	Class Service	MVP

Entity	Key fields	Relationships	Owned by	MVP or roadmap
Assignment	person_id, class_id, status (pending / confirmed / substitute requested)	Joins Person and Class	Class Service	MVP
Lesson note	class_id, author, content	Belongs to one Class	Class Service	Roadmap

Cognito owns credentials and login emails. Person covers teachers and leaders, so presidency emails go to people whose role is leader, and a leader can still be assigned to teach. A Person can have no account and no email: leaders still assign them and record their responses after talking to them. When someone signs up, `linkAccount()` attaches the account to the Person with the same email, or creates a new teacher.

On the schedule, a class is **covered** when it has `num_teachers` assignments that are all confirmed, **tentative** when it has `num_teachers` assignments and at least one is pending, and **uncovered** otherwise (too few assignments, or a substitute requested).



6. Technology choices and alternatives considered

Decision: put a Node backend between the UI and AWS

- **Chosen:** A Node (TypeScript) API on one EC2 instance behind Caddy. Only the API calls DynamoDB and SES.
- **Alternative considered:** No backend. The browser gets temporary AWS credentials from a Cognito identity pool and calls DynamoDB and SES directly, limited by fine-grained IAM policies.
- **Why the chosen option wins, given our goals:** Our rules are about roles and fields (a teacher may change only the status of their own assignment; a leader may change any), which are awkward to express and test as IAM conditions. IAM also can't limit SES to notification emails, so any signed-in user could send any email from our address. A backend keeps every rule in testable code, and it's what the proposal planned. The cost is a server to run, patch and pay for, and a single point of failure (§7).
- **Evidence:** CP3 spike (`aws-spike`): an Express/TypeScript app deployed to EC2 sent email through SES.
- **What would change our mind:** The instance's cost or upkeep becomes a problem. The same routes could move to Lambda behind API Gateway, keeping the checks on the server.

Decision: send notifications by email through AWS SES

- **Chosen:** Email through SES.

- **Alternative considered:** SMS through AWS SNS, which teachers would read fastest on a Sunday morning.
- **Why the chosen option wins, given our goals:** The 5-minute notification goals need a channel that works this semester. SMS needs registration steps we can't count on finishing in time. SES shares the stack's AWS account and is effectively free at our volume.
- **Evidence:** CP3 spike (`aws-spike`): SES delivered; SNS SMS did not.
- **What would change our mind:** Teachers say they don't check email fast enough to meet the 5-minute goal. SMS would then become a roadmap item.

Decision: store data in DynamoDB

- **Chosen:** Amazon DynamoDB.
- **Alternative considered:** PostgreSQL on the EC2 instance.
- **Why the chosen option wins, given our goals:** DynamoDB has no database to back up or upgrade, it keeps the EC2 instance stateless and rebuildable, and our volume (a few dozen people, a few hundred assignments a year) fits the free tier. PostgreSQL would tie the data to one machine and put backups on a solo developer. The cost is that our model is relational: instead of SQL joins, we design keys around our access patterns ("a teacher's assignments", "the next 4 Sundays") and enforce relationships in the services.
- **Evidence:** None yet for DynamoDB; the spike confirmed AWS access from this account.
- **What would change our mind:** The access patterns need joins or ad-hoc queries that don't fit the key design, or roadmap features make the model much more relational.

Decision: use Amazon Cognito for sign-in

- **Chosen:** A Cognito user pool for sign-up, sign-in and account recovery. The API verifies its JWTs.
- **Alternative considered:** Our own auth: hashed passwords in DynamoDB and our own session tokens.
- **Why the chosen option wins, given our goals:** The app stores ward members' contact details, so auth mistakes are the costliest kind, and a solo developer with 20–25 hours can't carefully build password storage, token expiry and reset flows. Cognito provides them within the free tier. The cost is lock-in to its flows and harder-to-debug configuration.
- **Evidence:** None yet. The proposal budgets about 4 hours for database and Cognito setup.
- **What would change our mind:** Setup runs well past that budget in milestone 1, or the verification step loses teachers who aren't comfortable with technology.

7. Risks and open questions

Risk or question	Why it matters	Owner	How we'll find out, and by when
The single EC2 instance goes down during the pilot.	Nobody can sign in or respond on a Sunday morning.	Hyrum	Script the setup in CP7 so it can be rebuilt; deploy in milestone 1 so it runs for three weeks before the pilot.
SES sandbox only sends to verified addresses.	Real teachers get no email until AWS grants production access.	Hyrum	Request production access in milestone 2; needed before the pilot.
Two people edit the same class at once.	A lost update could show a class as covered when it	Hyrum	Conditional writes; test with two browsers in milestone 4.

Risk or question	Why it matters	Owner	How we'll find out, and by when
	isn't.		
DynamoDB keys for relational data.	A poor key design makes the 4-Sunday view slow or awkward.	Hyrum	Write down the access patterns and prototype the tables in milestone 1.
Collecting contact information ethically.	Church member data has usage policies.	Hyrum	Check Member Tools policies and include only people who agree, before the pilot.
Will teachers use this rather than texts?	The value depends on teachers responding in the app.	Hyrum	Interview 1–2 other teachers before milestone 3.

8. Implementation plan

The build window is Weeks 9–13 (Oct 26 – Nov 29), about 20–25 hours. Milestones 1–4 are the proposal's P0 deliverable; 1–5 come to about 22 hours. Milestone 6 is a P1 stretch goal.

Milestone	What works when it's done	Depends on
1 · Nov 8 · ~7 h	**Thinnest slice, deployed:** the app runs on EC2 behind Caddy. A teacher logs in, sees their hand-seeded assignments and confirms one; the API rejects changes to others' assignments; the UI shows success or the DB-down error.	—
2 · Nov 15 · ~3 h	**Notifications:** confirming or requesting a substitute emails the presidency, with the email-failed warning. SES production access requested.	1
3 · Nov 18 · ~6 h	**Leader schedule:** a leader creates classes, assigns and reassigns teachers (emailing them), and sees the next 4 Sundays marked covered, tentative or uncovered.	1, 2
4 · Nov 22 · ~4 h	**Teachers outside the app, safe concurrent edits:** leaders add people without accounts or email and record their responses; sign-up links to an existing Person; simultaneous edits can't overwrite each other.	3
5 · Nov 29 and Dec 6 · ~2 h	**Ward pilot:** SES production access approved; the ward uses the app for two Sundays (the proposal's success criterion).	4
6 · stretch · ~3–4 h	**Admin and broadcast:** an admin creates accounts and sets roles; a leader messages all teachers at once.	2, 3

Sign-off

Reviewer	Role	Date	Approved
[name]	[role]	[date]	☐